



Storm Intensity Distribution — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 22-March-2026

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1 Test Results — Storm Intensity Distribution (MKM-SI-001)

Tests run on **22 March 2026 at 20:17:28**.

Table 1: Test Results Summary — Storm Intensity Distribution (MKM-SI-001)

Total Tests	105
Passed	105
Failed	0
Skipped	0
Duration	0.24s

Table 2: Detailed Test Results — Storm Intensity Distribution

Test Description	Acceptance Criteria	Result	Time
Custom parameters	<code>d.params.base_mean == 50;</code> <code>d.params.tail_threshold == 70</code>	Pass	0.000s
Default parameters	<code>d.params.base_mean == 45.0;</code> <code>d.params.base_std == 15.0 (+4 more)</code>	Pass	0.000s
From params object	<code>d.params.base_mean == 55</code>	Pass	0.000s
From scenario baseline	<code>d is not None;</code> <code>d.params.name ==</code> <code>"baseline"</code>	Pass	0.000s
From scenario unknown raises	—	Pass	0.000s
Seed gives deterministic output	<code>s1 == s2</code>	Pass	0.000s
Tail prob between 0 and 1	<code>0.0 < d._tail_prob < 1.0</code>	Pass	0.000s
All categories defined	<code>set(DURATION_PARAMS.keys()) ==</code> <code>expected</code>	Pass	0.000s
Catastrophic extended	<code>high == 240</code>	Pass	0.000s
Each tier's max roughly doubles the previous.	<code>1.5 <= ratio <= 3.0</code>	Pass	0.000s
Extreme extended	<code>DURATION_PARAMS["extreme"] == (36,</code> <code>72, 144)</code>	Pass	0.000s
Moderate extended	<code>low == 12; mode == 30 (+1 more)</code>	Pass	0.000s
Severe extended	<code>DURATION_PARAMS["severe"] == (24, 48,</code> <code>96)</code>	Pass	0.000s
Gap types	<code>GapType.SHORT.value == "short";</code> <code>GapType.MEDIUM.value == "medium" (+1</code> <code>more)</code>	Pass	0.000s
Sequence gap mapping	<code>SEQUENCE_GAP_TYPE[SequenceType.DOUBLET]</code> <code>== GapType.MEDIUM;</code> <code>SEQUENCE_GAP_TYPE[SequenceType.CLUSTER]</code> <code>== GapType.MEDIUM (+2 more)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Sequence types	SequenceType.ISOLATED.value == "isolated"; SequenceType.DOUBLET.value == "doublet" (+2 more)	Pass	0.000s
Storm count ranges	SEQUENCE_TYPE.STORM.COUNTS[SequenceType.ISOLATED] == (1, 1); SEQUENCE_TYPE.STORM.COUNTS[SequenceType.DOUBLET] == (2, 2) (+2 more)	Pass	0.000s
Above max is zero	d.exceedance_probability(d.params.max_intensity + 10) == 0.0	Pass	0.000s
Above threshold uses pareto	0 < p < 1	Pass	0.000s
At max intensity is zero	d.exceedance_probability(d.params.max_intensity) == 0.0	Pass	0.000s
At min intensity is one	d.exceedance_probability(d.params.min_intensity) == 1.0	Pass	0.000s
At threshold continuous	abs(p_just_below - p_just_above) < 0.05	Pass	0.000s
Below min is one	d.exceedance_probability(d.params.min_intensity - 5) == 1.0	Pass	0.000s
Below threshold uses normal	0 < p < 1	Pass	0.000s
Monotonically decreasing	all(probs[i] > probs[i + 1] for i in range(len(probs) - 1))	Pass	0.000s
Constants	EVENT_WINDOW_HOURS == 168; MIN_DRAINAGE_WINDOW_HOURS == 12 (+1 more)	Pass	0.000s
Exact boundary	fits_in_window([78.0, 78.0], [0.0])	Pass	0.000s
Exceeds window	not fits_in_window([72.0, 72.0], [48.0])	Pass	0.000s
Simple doublet fits	fits_in_window([24.0, 24.0], [48.0])	Pass	0.000s
Single storm	fits_in_window([100.0], [])	Pass	0.000s
Long gaps	GAP_PARAMS[GapType.LONG] == (48, 96, 144)	Pass	0.000s
Medium gaps	GAP_PARAMS[GapType.MEDIUM] == (24, 48, 72)	Pass	0.000s
Short gaps	GAP_PARAMS[GapType.SHORT] == (6, 18, 36)	Pass	0.000s
Doublet medium	get_gap_range(SequenceType.DOUBLET) == (24, 48, 72)	Pass	0.000s
Isolated zero	get_gap_range(SequenceType.ISOLATED) == (0, 0, 0)	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Get duration range	<code>get_duration_range("moderate") == (12, 30, 60)</code>	Pass	0.000s
Get max duration	<code>get_max_duration("catastrophic") == 240; get_max_duration("minimal") == 12</code>	Pass	0.000s
Ids unique	<code>len(ids) == 100</code>	Pass	0.000s
Sequence id format	<code>sid.startswith("STORM-"); len(sid) == 14</code>	Pass	0.000s
Sequence ids unique	<code>len(ids) == 100</code>	Pass	0.000s
Storm id format	<code>sid.startswith("STORM-"); len(sid) == 14</code>	Pass	0.000s
One prob returns min	<code>d.inverse_exceedance(1) == d.params.min_intensity</code>	Pass	0.000s
Result in bounds	<code>d.params.min_intensity <= v <= d.params.max_intensity; d.params.min_intensity <= v <= d.params.max_intensity</code>	Pass	0.000s
Roundtrip above threshold	<code>abs(recovered - target) < 3.0</code>	Pass	0.000s
Roundtrip below threshold	<code>abs(recovered - target) < 2.0</code>	Pass	0.000s
Zero prob returns max	<code>d.inverse_exceedance(0) == d.params.max_intensity</code>	Pass	0.000s
Fat tail higher exceedance	<code>p_fat > p_thin</code>	Pass	0.000s
Pareto above threshold between 0 and 1	<code>0 < p < 1</code>	Pass	0.000s
Pareto at threshold is one	<code>d.pareto_exceedance(d.params.tail_threshold) == 1.0</code>	Pass	0.000s
Pareto below threshold is one	<code>d.pareto_exceedance(d.params.tail_threshold - 5) == 1.0</code>	Pass	0.000s
100yr greater than 10yr	<code>i100 > i10</code>	Pass	0.000s
Ordering 10 50 100	<code>i10 < i50 < i100</code>	Pass	0.000s
Result in bounds	<code>d.params.min_intensity <= v <= d.params.max_intensity; d.params.min_intensity <= v <= d.params.max_intensity</code>	Pass	0.000s
Storms per year affects result	<code>many > few</code>	Pass	0.000s
Invalid category	—	Pass	0.000s
Mode should be near the most frequent value in a large sample.	<code>abs(median - mode) < 15</code>	Pass	0.005s
Line 50: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.	<code>low <= d <= high</code>	Pass	0.000s
Reproducibility	<code>d1 == d2</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sampled durations fall within [min, max] for the category. [category='baseline']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='catastrophic']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='extreme']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='minimal']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='moderate']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='severe']	<code>low <= d <= high</code>	Pass	0.000s
Doublet gaps should fall in medium range (24-72h).	<code>all(24 <= g <= 72 for g in gaps)</code>	Pass	0.000s
Isolated returns zero	<code>sample_gap(SequenceType.ISOLATED) == 0.0</code>	Pass	0.000s
Line 49: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.	<code>low <= g <= high</code>	Pass	0.000s
Persistent gaps should fall in short range (6-36h).	<code>all(6 <= g <= 36 for g in gaps)</code>	Pass	0.000s
Reproducibility	<code>g1 == g2</code>	Pass	0.000s
Within bounds[sequencetype.cluster] [seq_type=iSequenceType.CLUSTER: 'cluster']	<code>low <= g <= high</code>	Pass	0.000s
Within bounds[sequencetype.doublet] [seq_type=iSequenceType.DOUBLET: 'doublet']	<code>low <= g <= high</code>	Pass	0.000s
Within bounds[sequencetype.persistent] [seq_type=iSequenceType.PERSISTENT: 'persistent']	<code>low <= g <= high</code>	Pass	0.000s
Different seeds different output	<code>d1.sample(10) != d2.sample(10)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Sample n returns correct length	<code>len(samples) == 100</code>	Pass	0.000s
Sample single	<code>d.params.min_intensity <= v <=</code> <code>d.params.max_intensity</code>	Pass	0.000s
Samples within bounds	<code>all(d.params.min_intensity <= s <=</code> <code>d.params.max_intensity...</code>	Pass	0.001s
Seeded reproducibility	<code>s1 == s2</code>	Pass	0.000s
Create basic	<code>storm.storm_id == "STORM-abc12345";</code> <code>storm.duration.hours == 24.0</code>	Pass	0.000s
From dict round trip	<code>restored.storm_id ==</code> <code>original.storm_id;</code> <code>restored.duration.hours ==</code> <code>original.duration.hours (+6 more)</code>	Pass	0.000s
Precipitation rounding	<code>d["precipitation_mm"] == 35.12;</code> <code>d["intensity_factor"] == 1.234568 (+1</code> <code>more)</code>	Pass	0.000s
To dict	<code>d["storm_id"] == "STORM-test1234";</code> <code>d["start_time.hours"] == 48.0 (+6</code> <code>more)</code>	Pass	0.000s
Describe returns string	<code>isinstance(desc, str); "base_mean" in</code> <code>desc.lower() or "Base mean" in desc</code>	Pass	0.073s
Params roundtrip	<code>recovered.base_mean == 52;</code> <code>recovered.name == "custom"</code>	Pass	0.000s
Tail probability matches internal	<code>d.to_dict()['tail_probability'] ==</code> <code>pytest.approx(d._tail...</code>	Pass	0.000s
To dict keys	<code>'parameters' in result;</code> <code>'tail_probability' in result</code>	Pass	0.000s
Antecedent defaults	<code>seq.antecedent.soil_moisture ==</code> <code>"normal"; seq.antecedent.groundwater</code> <code>== "normal"</code>	Pass	0.000s
Create doublet	<code>seq.num_storms == 2; len(seq.storms)</code> <code>== 2 (+2 more)</code>	Pass	0.000s
From dict round trip	<code>restored.storm_id ==</code> <code>original.storm_id;</code> <code>restored.duration.hours ==</code> <code>original.duration.hours (+6 more)</code>	Pass	0.000s
Isolated no gaps	<code>len(seq.inter_storm_gaps_hours) == 0</code>	Pass	0.000s
Spatial correlation defaults	<code>seq.spatial_correlation_enabled</code> <code>is False;</code> <code>seq.spatial_correlation_range_km</code> <code>is None</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Spatial correlation enabled	<code>d["spatial.correlation.enabled"] is True; d["spatial.correlation.range_km"] == 43.2</code>	Pass	0.000s
To dict	<code>d["storm_id"] == "STORM-test1234"; d["start_time.hours"] == 48.0</code> (+6 more)	Pass	0.000s
Exceedance probs decreasing	<code>stats['prob.exceed_60'] >= stats['prob.exceed_70'] >= ...</code>	Pass	0.000s
Fat tail higher p99 than thin	<code>fat['p99'] > thin['p99']</code>	Pass	0.146s
Mean in reasonable range	<code>30 < stats['mean'] < 70</code>	Pass	0.000s
Percentiles ordered	<code>stats['p10'] <= stats['p25'] <= stats['p50'] <= \{\}</code> ...	Pass	0.000s
Required keys present	<code>k in stats</code>	Pass	0.000s
Duration mismatch	<code>any("total.duration.hours" in e for e in errors)</code>	Pass	0.000s
Exceeds precip window	<code>any("MAX_PRECIP_END_HOUR" in e for e in errors)</code>	Pass	0.000s
Invalid intensity category	<code>any("invalid intensity" in e for e in errors)</code>	Pass	0.000s
Invalid sequence type	<code>any("Invalid sequence.type" in e for e in errors)</code>	Pass	0.000s
Overlapping storms	<code>any("starts at" in e and "before storm" in e for e in err...</code>	Pass	0.000s
Storm count mismatch	<code>any("num.storms" in e for e in errors)</code>	Pass	0.000s
Too many storms	<code>any("not in [1, 5]" in e for e in errors)</code>	Pass	0.000s
Valid doublet	<code>errors == []</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (12 tests)	D. Kelly
07-Mar-2026	1.1	73 test(s) added; 12 test(s) removed (total: 146)	D. Kelly
08-Mar-2026	1.2	42 test(s) added (total: 115)	D. Kelly
12-Mar-2026	1.3	2 test(s) added (total: 117)	D. Kelly
22-Mar-2026	1.4	12 test(s) removed (total: 105)	D. Kelly